

Application of Persistent Homology in Forecasting Realized Volatility

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Abstract

This paper proposes a novel algorithm called Persistent Homology for Realized Volatility (PH-RV), which aims to effectively incorporate persistent homology (PH) into neural network models to increase their forecast accuracy in predicting realized volatility (RV). This paper also proposes a novel neural network model for multi-step forecasting that systematically yields statistically significantly more accurate forecasts than other forecasting models. The proposed model coined as NBEATSx-VP utilizes neural basis expansion analysis with exogenous variables (NBEATSx) network, and VIX close prices and 2-dimensional Wasserstein distances of stock log returns as exogenous variables. While considering Root-Mean-Square Error (RMSE), NBEATSx-VP demonstrates a minimum 9% improvement in forecast precision compared to benchmark models. Considering Quasi-likelihood (QLIKE), on the other hand, NBEATSx-VP achieves at least 13% more accurate forecasts than state-of-the-art models.

Keywords and JEL Codes: Neural Networks (C45), NBEATSx (C45), Realized Volatility Forecasting (C53).

1. Introduction

Topological data analysis (TDA) encompasses a growing set of methods that explore the topology, also commonly referred to as geometry, of data and leverage this topological information to infer patterns from the data. Since its creation, the field of TDA has demonstrated considerable potential in a wide variety of domains, particularly in those that handle high-dimensional data (Bubenik, 2015; Carlsson, 2009; Wasserman, 2017). TDA's success ranges from medical research (McInemey & Terzopoulos 1999; Nicolau et al., 2011) to financial research (Baitinger & Flegel, 2021; Gidea & Katz, 2018; Ismail et al., 2022). It is noteworthy that the majority of TDA literature utilizes the technique known as Persistent Homology (PH). PH is a remarkably effective algorithm that attempts to quantitatively find elementary persistent topological structures, patterns and qualitative features in data that are independent of coordinate representations and deformations related to noise (Baitinger & Flegel, 2021). By employing PH, researchers can uncover hidden yet fundamental patterns in data, leading to novel discoveries about patterns in the data (Bubenik, 2015; Carlsson, 2009; Wasserman, 2017).

While numerous fields have extensively researched the applications of TDA since the beginning of the 21st century, the finance literature had its first TDA research paper only in 2017 (Gidea, 2017). However, since then, a substantial amount of financial literature focusing on TDA has been produced focused on topics ranging from corporate and systemic financial stability (Gidea, 2017; Gidea & Katz, 2018; Ismail et al., 2022; Qiu et al, 2020) to investment strategies (Baitinger & Flegel, 2021; Goel et al., 2020).

Nonetheless, despite the major importance of modelling and forecasting stock realized volatility (RV) for investors and financial institutions (Albulescu, 2021; Atkins et al. 2018; Jebran et al., 2017; Wong et al; 2016), there currently exists no financial literature exploring TDA's possible applications in this area. Therefore, this paper has the objective to partially close this gap in the scientific community by proposing the employment of PH in forecasting RV for multi-step

forecasting ($H=5$). $H=5$ is chosen to reflect the prediction of a whole business week each time. The main contributions of this paper to the scientific community include:

- (i) **Proposition of a novel algorithm to measure topological chances of stock time-series:** this research modifies the topological change metric PH-based turbulence index (PHTI), originally proposed by Baitinger and Flegel (2021), to adapt it to be effectively employed in the prediction of RV.
- (ii) **Proposition of a novel and superior forecasting model:** this paper incorporates the proposed topological change metric into neural basis expansion analysis with exogenous variables (NBEATSx) model. This study demonstrates that the proposed model achieves statistically significantly more accurate forecasts than the existing state-of-the-art models in various scenarios for multi-step time series forecasting.
- (iii) **Provision of open-source code for the proposed algorithm and forecasting model:** this study supplies open-source code for an easy implementation of the proposed algorithm and forecasting model by researchers and practitioners.

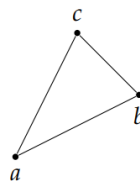
2. Persistent Homology and Proposed Algorithm

2.1 Persistent Homology

To ensure sparsity, this paper will solely provide a technical and mathematical explanation about PH and the proposed algorithm to measure topological chances of stock time-series for RV forecasting. Therefore, Some prior knowledge of the reader about topology and group theory is assumed. For an intuitive and conceptual explanation about PH, please see Baitinger and Flegel (2021), and Ismail et al. (2022).

Before introducing PH, Betti numbers must be explained. The d^{th} Betti number, β_d , is essentially the number of d -dimensional holes in a certain topological shape, also called as simplicial complex (SC). For instance, β_0 can be thought as the number of connected components (geometric elements), whereas β_1 as the number of (one-dimensional) holes and β_2 as the number of 3-d spaces that are encapsulated (voids). The elements of a SC, on the other hand, are called simplices, and all SCs can be considered a group. For instance, the SC in Figure 1 can be written as $K = \{\{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}\}$.

Figure 1 Triangle



Graphically, it is obvious that $\beta_0=1$ (one triangle), $\beta_1=1$ (one one-dimensional hole), and $\beta_2=0$ (zero voids). However, as the number of dimensions of a SC increases, it is no longer possible to

visually determine its Betti numbers. Thus, one can use the following equations to mathematically estimate β_d :

$$\beta_d = \text{rank } H_d \quad (1)$$

$$H_d = \ker \partial_d / \text{im } \partial_{d+1} \quad (2)$$

$$\partial_d(K) = \sum_m \{v_0, \dots, \widehat{v_m}, \dots, v_d\} \quad (3)$$

where, ∂_d is the d^{th} boundary homomorphism function applied to K , $\ker$ is the kernel of a homomorphism and im is the image of a homomorphism.

Applying these formulas to K to determine β_0 , we get:

$$\partial_0(K) = \{0, 0, 0\}$$

$$\partial_1(K) = \{\{a\} + \{b\}, \{b\} + \{c\}, \{a\} + \{c\}\}$$

$$\therefore \ker \partial_0(K) = \text{span}(\{a\}, \{b\}, \{c\}) = (\mathbb{Z}/2\mathbb{Z})^3$$

$$\text{im } \partial_1(K) = \text{span}(\{a\} + \{b\}, \{b\} + \{c\}) = (\mathbb{Z}/2\mathbb{Z})^2$$

$$\therefore H_0(K) = (\mathbb{Z}/2\mathbb{Z})^3 / (\mathbb{Z}/2\mathbb{Z})^2 = \mathbb{Z}/2\mathbb{Z}$$

$$\beta_0(K) = \text{rank } H_0(K) = 1$$

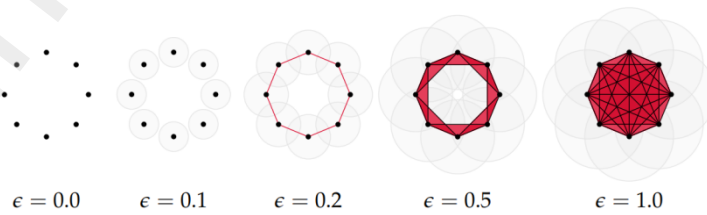
where, $\mathbb{Z}/2\mathbb{Z}$ is the cyclic group of order 2 and $\text{span}(\{a\} + \{b\}, \{b\} + \{c\}) = \text{span}(\{a\} + \{b\}, \{b\} + \{c\}, \{a\} + \{c\})$ since $\{a\} + \{c\} = \{a\} + \{b\} + \{b\} + \{c\}$ in $\mathbb{Z}/2\mathbb{Z}$.

Now consider the basket of stocks $S_t = \{s_t^1, \dots, s_t^n\}$ at time t and a distance metric dist . By picking a threshold ϵ , one can build the Vietoris-Rips (VR) complex $\mathcal{V}_\epsilon$ defined as:

$$\mathcal{V}_\epsilon(S_t) := \{\sigma \subseteq S_t \mid \forall u, v \in \sigma: \text{dist}(u, v) \leq \epsilon\} \quad (4)$$

In other words, $\mathcal{V}_\epsilon$ contains all simplices whose diameter is less than or equal to ϵ . Figure 2 shows VR complexes of an arbitrary dataset for five different thresholds.

Figure 2 Vietoris-Rips complexes for various thresholds

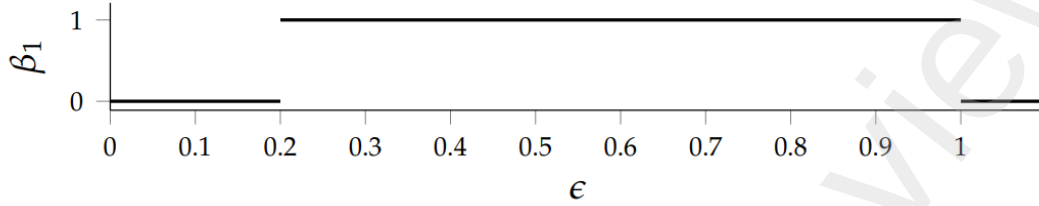


Considering a range of N thresholds $\epsilon = \{\epsilon_1, \dots, \epsilon_N\}$, where $\epsilon_1 \leq \epsilon_2 \leq \dots \leq \epsilon_N$, the following holds:

$$\mathcal{V}_{\epsilon_1}(S_t) \subseteq \mathcal{V}_{\epsilon_2}(S_t) \subseteq \dots \subseteq \mathcal{V}_{\epsilon_N}(S_t)$$

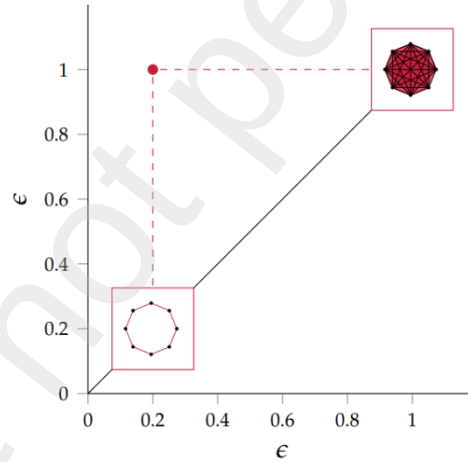
Therefore, Figure 3 can be constructed by estimating $\beta_1(\mathcal{V}_{\epsilon_i}) \forall i \in \{1, \dots, N\}$ for the VR complexes of Figure 2.

Figure 3 1st Betti number for each threshold



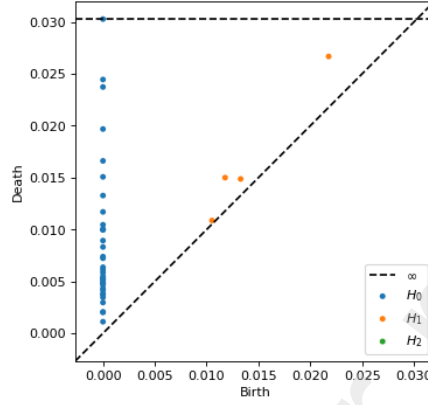
Creation in ϵ_i can be defined as $c \in \beta_d(\mathcal{V}_{\epsilon_i})$, but $c \notin \beta_d(\mathcal{V}_{\epsilon_{i-1}})$, and destruction in ϵ_i as $c \notin \beta_d(\mathcal{V}_{\epsilon_i})$, but $c \in \beta_d(\mathcal{V}_{\epsilon_{i-1}})$, where $c := \{0, 1, \dots, C\}$ and $\beta_d(\mathcal{V}_{\epsilon_i}) := \{0, 1, \dots, \beta_d\}$. Hence, in Figure 3, $c = \{0, 1\}$, and there is a creation in $\epsilon_i = 0.2$ and a destruction in $\epsilon_i = 1$. By accounting for the creation and destruction thresholds for each c , the so-called Persistence Homology Diagram (PD) can be created:

Figure 4 Persistence Homology Diagram for β_1



A real-life example of a PD that is used in this research is depicted in Figure 5. Figure 5 shows the PD of the log returns of the basket of stocks composed of Standard & Poor's 500 (S&P 500), Dow Jones Industrial Average (DJIA) and Russell 2000 (RUT) from 1st of February, 2000 until 21 March, 2000. The H_0 points can be ignored since they represent single data points that needless to say do not show any important topological structure. Yet, while H_1 points close to the diagonal are simply topological structures created by noise (Bubenik, 2015; Carlsson, 2009; Wasserman, 2017), the H_1 points far from the diagonal can show that the data has a (or some) topological structure(s) that ought to be exploited.

Figure 5 PD of S&P 500, DJIA, and RUT log returns



The last mathematical concept to be presented is the Wasserstein distance defined by Equation 5. This metric is used to estimate how different two PDs are from each other, and thus calculate the topological difference between two datasets or time frames.

$$W_d(PD_1, PD_2) := \left(\inf_{\eta: PD_1 \rightarrow PD_2} \sum_{x \in PD_1} \|x - \eta(x)\|_\infty^d \right)^{\frac{1}{d}} \quad (5)$$

where, $\inf$ is the infimum operator, η is an arbitrary bijection (function) mapping elements of PD_1 onto PD_2 , and d is the considered dimension.

In simple terms, PH can be summarized as the procedure of finding topological features of data by 1. plotting its datapoints, 2. surrounding the datapoints by discs or spheres of radius ϵ and connecting datapoints if the respective datapoints circles touch each other, 3. estimating the number of holes for d dimensions with Betti numbers, 4. repeating this process for various values of ϵ , and finally 5. creating a PD with the creation and destruction ϵ values for each Betti number in each dimension. Afterwards, it is possible to compare the topological features of two datasets or time frames by comparing their respective PDs with each other through Wasserstein distance.

2.2 Proposed Algorithm

The proposed algorithm, coined as Persistent Homology for Realized Volatility (PH-RV), can be summarized as:

1. Construct a sparse basket of stocks containing stocks that are logically similar to each other, e.g. same industry, same economic region, close competitors, etc.
2. For the basket of stocks $S_t = \{s_t^1, \dots, s_t^n\}$ ranging from t_0 to T , estimate the log-return of each stock $R_t = \{r_t^1, \dots, r_t^n\}$ at each time step and m , which is equal to $T - 41$.

3. Estimate m numbers of PDs, each limited to a maximum two dimensions for the Betti numbers, by using the dataset $= \{R_{t-20}, \dots, R_t\}$ for each considered time step. That is, estimate PDs for m days using last month's log returns of each stock in the stock basket.
4. For m days, estimate the 2-dimensional Wasserstein distance between PD_t and PD_{t-1} .
5. Utilize the 2-dimensional Wasserstein distances as an exogenous variable for the chosen RV forecasting model.

The Python open-source code provided by this research for this algorithm can be found on <https://github.com/hugogobato/Application-of-Persistent-Homology-in-Forecasting-Realized-Volatility/tree/57e0b4dca29fe82f002d629b76322d9517d6e9ff/PH-RV%20algorithm>. This open-source code employs the library devised by Tralie et al. (2018) and allows the utilization of Yahoo Finance's API for fast extraction of stock data.

3. Methodology

3.1 Sample

For the main sample, three famous stock indexes are used, namely Standard & Poor's 500 (S&P 500), Dow Jones Industrial Average (DJIA) and Russell 2000 (RUT). The sample period for these stocks ranges from 1st of January, 2000 until 30 March, 2022. However, after employing PH-RV, this period ranges from 1st of February, 2000 until 2 March, 2022, encompassing a total of 5,557 daily observations. The same sample is used for two out of the three robustness tests discussed in Section 3.6. The third robustness test, on the other hand, employs the following three European stock indexes: Financial Times Stock Exchange 100 (FTSE), DAX Performance (GDAXI), and STXE 600 (STOXX). For these European indexes, the sample period after the application of PH-RV ranges from 25 May, 2004 until 2 March, 2022, a total of 4,390 daily observations. The sample period for the European indexes is a bit shorter due to data availability.

Further, both samples are divided into 80% training sample and 20% test sample. Moreover, while the training sample is used to estimate the optimal parameters of the studied models, 25% of the training sample is used as validation sample to determine the hyperparameters of the neural network models (see Section 3.4 for more details).

Lastly, all stock data in this research is retrieved from Compustat – Capital IQ dataset.

3.2 Realized Volatility Proxy

The RV proxy chosen for this paper is Yang & Zhang's RV (2000) due to its robustness to drift and consistence in dealing with opening price jumps (Yang & Zhang, 2000). Its choice is also motivated by its accessibility thanks to the fact that this proxy only needs High, Low, Open, and Close prices for its estimation (Yang & Zhang, 2000). Yang & Zhang's RV is defined as:

$$RV_t = \sqrt{\sigma_t^2} \quad (6)$$

$$\sigma_t^2 = \sigma_{O_t}^2 + k \sigma_{C_t}^2 + (1 - k) \sigma_{RS_t}^2 \quad (7)$$

with,

$$\sigma_{o_t}^2 = \frac{1}{n-1} \sum_{t=1}^T (o_t - \bar{o}) \quad (8)$$

$$\sigma_{c_t}^2 = \frac{1}{n-1} \sum_{t=1}^T (c_t - \bar{c}) \quad (9)$$

where, o_t = opening price at time t , $\bar{o}$ = opening price mean, c_t = close price at time t , $\bar{c}$ = close price mean, k = parameter, and $\sigma_{RS_t}^2$ = Rogers et al. (1994) variance estimation.

Yang & Zhang's (2000) empirical research indicates that the best k value is given as:

$$k = \frac{0.34}{1.34 + \frac{n+1}{n-1}} \quad (10)$$

Therefore, this k value is used in this research.

3.3 Considered Models

This paper examines two neural network models and two linear models. Given the task of multi-step forecasting for 5 time-steps ($H=5$), the choice of NBEATSx comes naturally since this neural network, with VIX close prices (VIX) as an exogenous variable, yields statistically significantly more accurate and robust forecasts for multi-step forecasting than other neural network models (Souto & Moradi, 2023). Therefore, the two neural network models are NBEATSx with VIX and NBEATSx with VIX and 2-d Wasserstein distances estimated by PH-RV. The first model is henceforth mentioned as NBEATSx-V while the latter model as NBEATSx-VP. Additionally, for a thorough explanation of NBEATSx architecture, see Olivares et al. (2022), and Souto and Moradi (2023).

Concerning the linear models, the standard benchmarks Heterogeneous Autoregressive (HAR) model and Heterogeneous Autoregressive with eXogenous variables (HARX) model are employed. The HAR and HARX models are respectively defined by Equation 11 and 12:

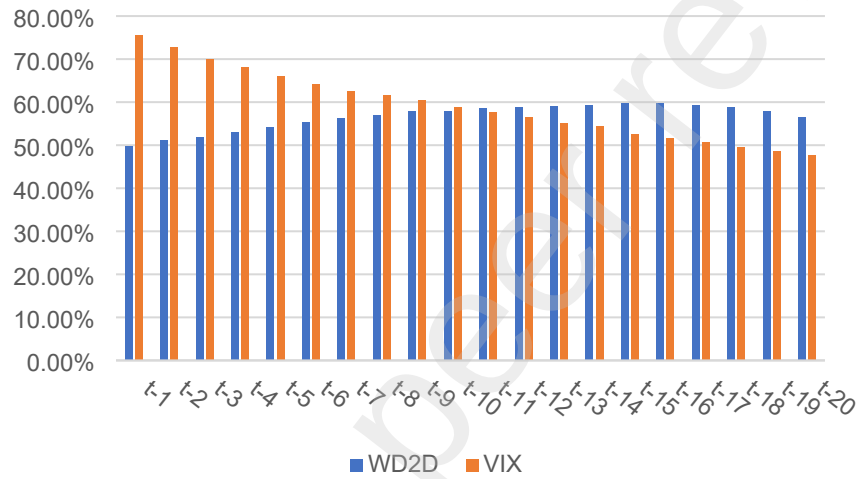
$$RV_t = \beta_0 + \beta_d RV_{t-1} + \beta_w \overline{RV}_{t-1:t-5} + \beta_m \overline{RV}_{t-1:t-22} \quad (11)$$

$$RV_t = \beta_0 + \beta_d RV_{t-1} + \beta_w \overline{RV}_{t-1:t-5} + \beta_m \overline{RV}_{t-1:t-22} + \beta_x \mathbf{X}_{t-1} \quad (12)$$

where, $\overline{RV}_{t-1:t-5}$ and $\overline{RV}_{t-1:t-22}$ are respectively the week and month's moving average of RV and $\mathbf{X}_{t-1}$ is the $1 \times N$ vector of lagged exogenous variables, with N being the number of considered exogenous variables. For this study, the HARX model encompasses VIX and 2-d Wasserstein distances as exogenous variables.

The choice for use of VIX and 2-d Wasserstein distances (WD2D henceforth) is motivated by the fact that for all studied stocks, while the correlation between RV_t and VIX_{t-i} diminishes as i increases, the correlation between RV_t and $WD2D_{t-i}$ increases as i increases, reaching its global maximum at $t - 15$. This pattern is illustrated in Figure 5. This behavior suggests that these two exogenous variables possess complementary characteristics; that is, VIX captures short-term patterns of RV, while WD2D captures long-term patterns of RV.

Figure 5 Correlation of Exogenous Variables with S&P 500 RV



3.4 Hyperparameters Search Space

Table 1 presents the hyperparameters search space utilized to determine the optimal hyperparameters for each studied stock index using the validation sample results.

Table 1 Hyperparameters Search Space

Hyperparameters	NBEATSx
Number of Lags	[5, 10, 21, 63, 84, 189, 252]
Scaler Type	[robust, standard, minmax]
Number of Epochs	[25, 50, 100, 150, 250, 350, 450, 550, 750]
Loss Functions	[MSE, MAE, MQLoss (level =[90]), MQLoss (level =[80, 90]), MQLoss (level =[95]), MQLoss (level =[75])]
Learning Rate	[5.00E-4, 1.00E-4, 5.00E-5, 1.00E-5]
Number of Learning Decays	[5, 3, 2, 1]
Number of units of each hidden layer	[[[712, 712], [712, 712]], [[512, 512], [512, 512]], [[250, 250], [250, 250]], [[100, 100], [100, 100]]]
Number of blocks per stack	[[1, 1], [2, 2], [3, 3], [5, 5]]

Stack Types	[[identity, identity, exogenous], [identity, trend, exogenous], [identity, seasonality, exogenous], [trend, seasonality, exogenous]]
Number of Harmonics	[0, 1]
Number of Polynomials	[0, 1]
Random Seed Range	[0, ... , 129228148]

A total of 65 trials are performed for each stock index to explore the hyperparameter search space, of which 20 trials are specially dedicated to exploring the random seed range.

3.5 Forecast Assessment

This paper employs two error metrics to assess the models' forecast accuracy in the out-sample data (test sample). These are namely Root-Mean-Square Error (RMSE) and Quasi-likelihood (QLIKE). The choice of RMSE is motivated by its easily interpretable results when compared to Mean-Square Error (MSE) while being more sensitive to errors further from the mean than Mean Absolute Error (MAE). QLIKE, on the other hand, is utilized to account for errors in a relative manner (i.e., QLIKE is a scale-free error measure). This property of QLIKE allows it to be easily used to compare model's performance in datasets with different scales. Concerning RV, QLIKE can be used to compare a model's accuracy performance among stocks of high and low volatility. The selected error measures are explained by Equation 13 and 14:

$$RMSE = \sqrt{\frac{\sum_{t=1}^T (RV_t - \widehat{RV}_t)^2}{T}} \quad (13)$$

$$QLIKE = \frac{\sum_{t=1}^T \left[\frac{RV_t}{\widehat{RV}_t} - \log \left(\frac{RV_t}{\widehat{RV}_t} \right) - 1 \right]}{T} \quad (14)$$

where, T is the number of forecasted time-steps and $\widehat{RV}_t$ is the forecasted RV_t . Additionally, since RV values are always positive, $\log\left(\frac{RV_t}{\widehat{RV}_t}\right)$ is always well defined.

Further, Diebold-Mariano (DM) tests are performed to determine whether the forecasts from a pair of models present statistically significantly different accuracy. DM tests were originally proposed by Diebold and Mariano (1995), and later modified by Harvey et al. (1997) to take serial dependence into account. Consequently, this paper employs the modified DM tests as proposed by Harvey et al. (1997) with MSE as the error metric. The following two hypotheses are used for DM tests:

- Null Hypothesis (H_0): there is no statistically significant difference between the forecasts of the considered pair of models.
- Alternative Hypothesis (H_1): there is a statistically significant difference between the forecasts of the considered pair of models.

In addition, Model Confidence Set (MCS) tests, proposed by Hansen et al. (2011), are also employed in this research. MCS tests identify a subset of models with significantly superior performance from model candidates, at a given level of confidence (a p -value threshold). For more details about this procedure, see Hansen et al (2011).

In this research, the used p -value thresholds are 0.01, 0.05, and 0.10, and they are respectively represented in the tables of Section 4 and 5 by ***, **, and *. To ensure sparsity, the p -values of DM tests are shown only for the pairs of NBEATSx and the other models. If the reader wants to see the DM tests results for other pair of models, please see <https://github.com/hugogobato/Application-of-Persistent-Homology-in-Forecasting-Realized-Volatility.git>.

Finally, graphical comparisons are performed for the forecasts of each stock index. However, to ensure sparsity, only the graphical comparison of S&P 500 for the main sample is displayed in this paper. To find the graphical comparisons for all stock indexes and robustness tests, please see <https://github.com/hugogobato/Application-of-Persistent-Homology-in-Forecasting-Realized-Volatility.git>.

3.6 Robustness Tests

Three aspects regarding the robustness of the proposed model are considered: (i) an alternative training set size for the main sample, (ii) similar multi-step forecast sizes ($H=4$ and $H=6$), and (iii) stocks from other geographic regions than the main sample.

For the first robustness test (i), the training sample size is reduced to half of its original size, while for the second robustness test (ii), the multi-step forecast sizes of $H=4$ and $H=6$ are considered. Finally, the third robustness test (iii) employs European stock indexes, namely FTSE, GDAXI, and STOXX.

4. Main Sample Results

Table 2 shows the chosen optimal hyperparameters of the neural network models for each index after the validation stage.

Table 2 Optimal Hyper parameters

Models	Hyperparameters		
	S&P 500	DJIA	RUT
NBEATSx-V	Number of lags: 63 Scaler Type: minmax Stack Types: [seasonality, identity, exogenous] Number of units of each hidden layer: [[712, 712],[712, 712]] Learning rate: 1.00E ⁻⁴ Loss Function: MQLoss (level =[90]) Number of blocks: [2, 2] Number of epochs: 250 Random seed: 67671771	Number of lags: 63 Scaler Type: robust Stack Types: [trend, seasonality, exogenous] Number of units of each hidden layer: [[512, 512],[512, 512]] Learning rate: 5.00E ⁻⁴ Loss Function: MQLoss (level =[75]) Number of blocks: [5, 5] Number of epochs: 100 Random seed: 83642934	Number of lags: 63 Scaler Type: minmax Stack Types: [trend, identity, exogenous] Number of units of each hidden layer: [[712, 712],[712, 712]] Learning rate: 1.00E ⁻⁵ Loss Function: MAE Number of blocks: [2, 2] Number of epochs: 750 Random seed: 22782732
NBEATSx-VP	Number of lags: 21 Scaler Type: minmax Stack Types: [trend, identity, exogenous] Number of units of each hidden layer: [[250, 250],[250, 250]] Learning rate: 1.00E ⁻⁴ Loss Function: MSE Number of blocks: [1, 1] Number of epochs: 550 Random seed: 69871046	Number of lags: 21 Scaler Type: standard Stack Types: [trend, seasonality, exogenous] Number of units of each hidden layer: [[100, 100],[100, 100]] Learning rate: 5.00E ⁻⁵ Loss Function: MAE Number of blocks: [3, 3] Number of epochs: 450 Random seed: 62584243	Number of lags: 5 Scaler Type: standard Stack Types: [trend, identity, exogenous] Number of units of each hidden layer: [[512, 512],[512, 512]] Learning rate: 1.00E ⁻⁵ Loss Function: MQLoss (level =[95]) Number of blocks: [1, 1] Number of epochs: 450 Random seed: 87819447

Table 3, on the other hand, presents the error measures and *p*-values of the statistical tests for each index. Notably, NBEATSx-VP is systematically superior to all other studied models considering both RMSE and QLIKE. Concerning RMSE, NBEATSx-VP respectively achieves at least 9%, 10%, and 10% better results than NBEATSx-V, HAR, and HARX. Regarding QLIKE, on the other hand, NBEATSx-VP respectively yields at least 13%, 18%, and 19% more accurate forecasts than NBEATSx-V, HAR, and HARX. This superiority is also reflected in the *p*-values of DM and MCS tests. Apart from three exceptions (HAR for S&P 500 and DJIA, and HARX for S&P 500), all *p*-values are below the conservative threshold of 0.01. Therefore, it can be concluded that NBEATSx-VP yields statistically significantly more accurate forecasts than all other models for the main sample.

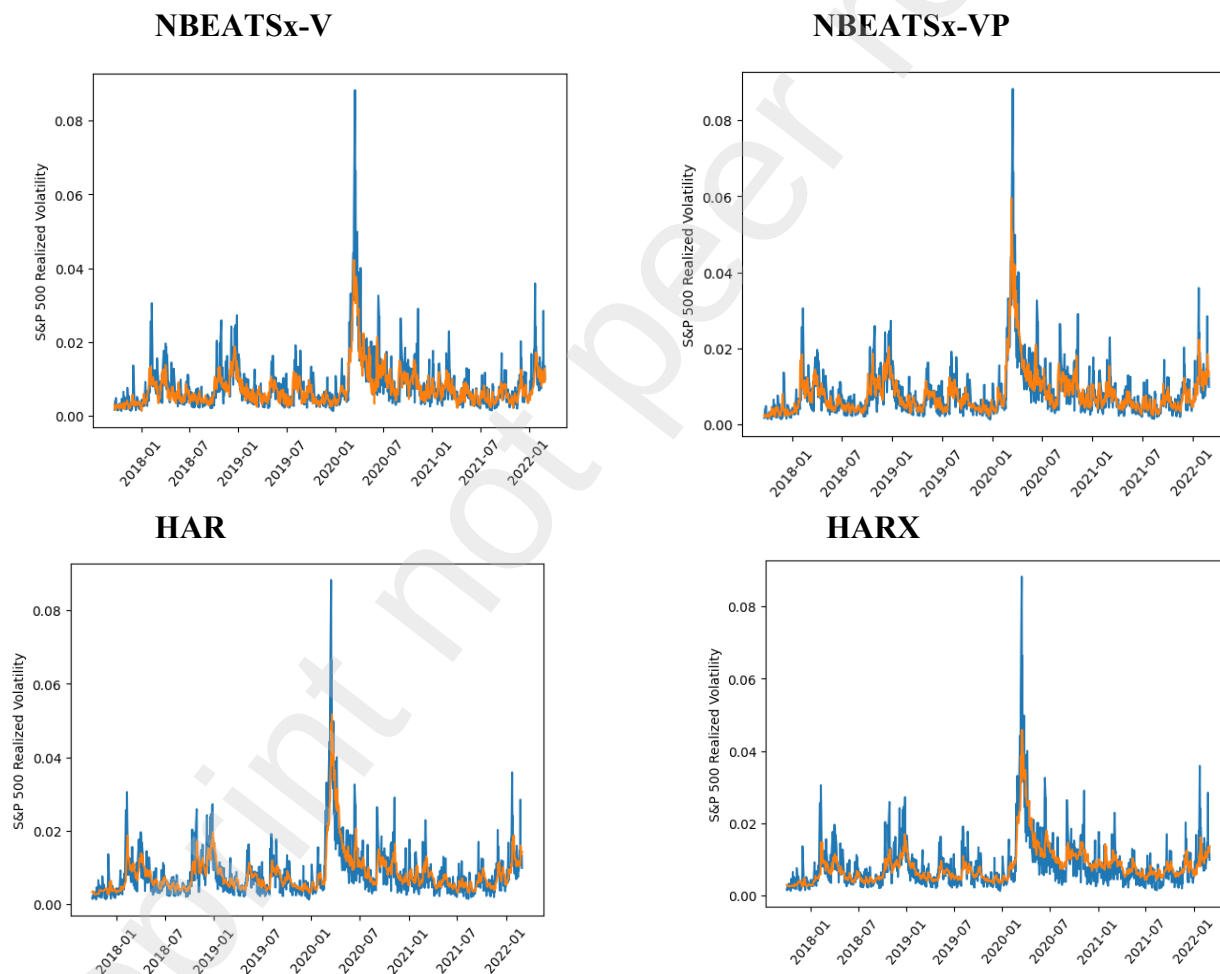
Table 3 Main Sample Results

Stock Index	Measures	NBEATSx-V	NBEATSx-VP	HAR	HARX
S&P 500	RMSE	0.472%	0.418%	0.462%	0.466%
	QLIKE	9.81%	8.51%	10.33%	10.46%
	DM test	7.61E ^{-3***}	-	0.02**	0.05**
	MCS test	0.01***	1.00	0.01***	0.01***

DJIA	RMSE	0.516%	0.470%	0.528%	0.532%
	QLIKE	9.94%	8.39%	10.16%	10.32%
	DM test	0.01***	-	0.03**	2.28E ⁻⁹ ***
	MCS test	8.00E ⁻³ ***	1.00	1.00E ⁻³ ***	8.00E ⁻³ ***
RUT	RMSE	0.490%	0.429%	0.493%	0.482%
	QLIKE	10.00%	7.47%	9.30%	9.03%
	DM test	9.13E ⁻⁸ ***	-	3.54E ⁻⁶ ***	4.58E ⁻⁵ ***
	MCS test	9.00E ⁻³ ***	1.00	3.00E ⁻³ ***	9.00E ⁻³ ***

Figure 6 displays the graphical comparison for the forecasts for S&P 500. The results are similar to those in Table 3, namely the forecasts of NBEATSx-VP are considerably superior to the forecasts of the other model.

Figure 6 Graphical Comparison for S&P 500



Despite the remarkably positive results of NBEATSx-VP for the main sample, robustness tests are still required to conclude whether the superiority of NBEATSx-VP is due to randomness or an intrinsic advantage thanks to the addition of WD2D.

5. Robustness Tests

Once again, the robustness tests performed and discussed in this section are (i) an alternative training set size for the main sample, (ii) similar multi-step forecast sizes ($H=4$ and $H=6$), and (iii) stocks from other geographic regions than the main sample.

In Table 4, the results of the first robustness test are found. NBEATSx-VP is anew superior to all models considering all stock indexes and error measures. For this robustness test, NBEATSx-VP respectively produces at least 11%, 8%, and 11% more accurate forecasts than NBEATSx-V, HAR, and HARX considering RMSE. Concerning QLIKE, NBEATSx-VP respectively achieves at least 10%, 9%, and 18% better results than NBEATSx-V, HAR, and HARX. Again, the p -values of the statistical tests confirm that the forecasting superiority of NBEATSx-VP is statistically significant. Nonetheless, it is worth mentioning that now there are six cases in which the p -values are not below the conservative threshold of 0.01.

Table 4 First Robustness Test Results

Stock Index	Measures	NBEATSx-V	NBEATSx-VP	HAR	HARX
S&P 500	RMSE	0.491%	0.428%	0.465%	0.491%
	QLIKE	10.06%	8.67%	10.30%	11.12%
	DM test	0.01***	-	0.03**	0.03**
	MCS test	7.00E-3***	1.00	6.00E-3***	6.00E-3***
DJIA	RMSE	0.563%	0.481%	0.539%	0.573%
	QLIKE	10.42%	9.33%	10.28%	11.40%
	DM test	0.04**	-	3.08E-3***	9.30E-3***
	MCS test	0.02**	1.00	0.02**	0.03**
RUT	RMSE	0.482%	0.429%	0.496%	0.480%
	QLIKE	9.19%	7.40%	9.29%	8.98%
	DM test	4.87E-5***	-	6.66E-6***	1.15E-4***
	MCS test	0.01***	1.00	1.00E-4***	9.00E-3***

Table 5 presents the results of the second robustness test for both $H=4$ and $H=6$. Regarding $H=4$, NBEATSx-VP respectively yields at least 6%, 9% and 10% more accurate forecasts than NBEATSx-V, HAR, and HARX concerning RMSE. Regarding QLIKE, on the other hand, NBEATSx-VP respectively achieves at least 10%, 16%, and 18% better results than NBEATSx-V, HAR, and HARX. The statistical tests for $H=4$ have similar results to $H=5$; again, there are only three cases in which the p -values are not below the conservative threshold of 0.01.

Considering $H=6$, the superiority of NBEATSx-VP is less evident. Regarding RMSE, NBEATSx-VP respectively produces at least 2%, 3%, and 3% better results than NBEATSx-V, HAR, and HARX. Whereas concerning QLIKE, NBEATSx-VP respectively yields at least 9%, 8%, and 8% more accurate forecasts than NBEATSx-V, HAR, and HARX. The less present superiority of NBEATSx-VP is also seen in the statistical tests results. Regarding S&P 500, although all p -values of DM tests are below the threshold of 0.05, all p -values of MCS test are above the relaxed threshold of 0.10. Concerning DJIA, all p -values of both tests are above the threshold of 0.10, except for the p -value of MCS test between NBEATSx-VP and HARX. Considering RUT, on the other hand, all p -values of DM tests are below the threshold of 0.05, while only the p -value of MCS test between NBEATSx-VP and HARX is above the threshold of

0.10. Consequently, it could be hypothesized that the superiority of NBEATSx-VP over other forecasting models vanishes as H increases. Nevertheless, more research is needed to test such a hypothesis and find a scientific explanation for the phenomenon.

Table 5 Second Robustness Test Results

Stock Index	Measures	NBEATSx-V	NBEATSx-VP	HAR	HARX
$H=4$					
S&P 500	RMSE	0.459%	0.398%	0.462%	0.469%
	QLIKE	9.31%	8.37%	10.33%	10.65%
	DM test	1.92E ^{-3***}	-	1.61E ^{-4***}	1.92E ^{-3***}
	MCS test	0.03**	1.00	0.03**	0.03**
DJIA	RMSE	0.515%	0.482%	0.528%	0.533%
	QLIKE	9.54%	8.51%	10.16%	10.40%
	DM test	3.54E ^{-3***}	-	2.26E ^{-3***}	3.08E ^{-3***}
	MCS test	6.00E ^{-3***}	1.00	5.00E ^{-3***}	5.00E ^{-3***}
RUT	RMSE	0.480%	0.421%	0.493%	0.481%
	QLIKE	10.15%	7.30%	9.24%	8.97%
	DM test	1.35E ^{-9***}	-	3.09E ^{-6***}	2.44E ^{-5***}
	MCS test	1.00E ^{-4***}	1.00	1.00E ^{-3***}	5.00E ^{-3***}
$H=6$					
S&P 500	RMSE	0.468%	0.432%	0.461%	0.463%
	QLIKE	10.11%	9.17%	10.19%	10.24%
	DM test	0.02**	-	0.01***	4.62E ^{-7***}
	MCS test	0.15	1.00	0.15	0.15
DJIA	RMSE	0.525%	0.514%	0.528%	0.532%
	QLIKE	10.69%	9.35%	10.14%	10.21%
	DM test	0.48	-	0.23	0.16
	MCS test	0.11	1.00	0.11	0.04**
RUT	RMSE	0.494%	0.453%	0.493%	0.484%
	QLIKE	9.51%	8.13%	9.20%	8.97%
	DM test	1.18E ^{-3***}	-	5.44E ^{-3***}	0.03**
	MCS test	0.06*	1.00	0.02**	0.12

Finally, Table 6 shows the results of the third robustness test. Considering RMSE, NBEATSx-VP respectively yields at least 2%, 4%, and 5% more accurate results than NBEATSx-V, HAR, and HARX. Regarding QLIKE, on the other hand, NBEATSx-VP respectively achieves at least 4% and 6% better results than HAR, and HARX. Interestingly, NBEATSx-VP fails to yield more accurate forecasts than NBEATSx-V for STOXX concerning QLIKE. Nonetheless, the p -values of the statistical tests do show a superiority of NBEATSx-VP over all models. Apart from the statistical tests between NBEATSx-VP and NBEATSx-V for STOXX, all p -values are below the threshold of 0.05, with the majority being below the conservative threshold of 0.01.

Table 6 Third Robustness Test Results

Stock Index	Measures	NBEATSx-V	NBEATSx-VP	HAR	HARX
	RMSE	0.404%	0.383%	0.408%	0.423%

FTSE	QLIKE	8.09%	7.76%	8.24%	8.50%
	DM test	0.01***	-	9.46E-5***	7.14E-5***
	MCS test	0.02**	1.00	0.02**	0.02**
GDAXI	RMSE	0.571%	0.560%	0.597%	0.587%
	QLIKE	9.69%	9.46%	10.55%	10.09%
	DM test	0.01***	-	1.27E-3***	1.65E-3***
	MCS test	1.00E-3***	1.00	1.00E-4***	1.00E-4***
STOXX	RMSE	0.412%	0.397%	0.415%	0.435%
	QLIKE	9.48%	9.62%	10.02%	10.61%
	DM test	0.35	-	2.58E-3***	1.60E-3***
	MCS test	0.22	1.00	0.01***	0.01***

Hence, it can be confidently affirmed that NBEATSx-VP is statistically superior to the other studied models, showing another positive application of TDA in finance.

6. Conclusions

This paper proposes an algorithm, coined as Persistent Homology for Realized Volatility (PH-RV), which effectively employs 2-dimensional Wasserstein distances as an exogenous variable in neural network models to increase their forecast accuracy. This paper also proposes a novel neural network model for multi-step forecasting ($H=5$) that systematically yields statistically significantly more accurate forecasts than other forecasting models. The proposed model, NBEATSx-VP, utilizes the neural basis expansion analysis with exogenous variables (NBEATSx) network, and VIX close prices and 2-dimensional Wasserstein distances of stock log returns as exogenous variables.

The proposed model is compared with benchmark forecasting models through the use of two error measures, Root-Mean-Square Error (RMSE) and Quasi-likelihood (QLIKE), and two robust statistical tests, Diebold-Mariano (DM) tests and Model Confidence Set (MCS) tests. The benchmark forecasting models utilized in this study are namely Heterogeneous Autoregressive (HAR) model, Heterogeneous Autoregressive with eXogenous variables (HARX) model, and NBEATSx with VIX close prices as an exogenous variable (NBEATSx-V).

Incidentally, the main sample of this study ranges from 1st of February, 2000 until 2 March, 2022 and encompasses daily realized volatility (RV) observation of Standard & Poor's 500 (S&P 500), Dow Jones Industrial Average (DJIA) and Russell 2000 (RUT). Moreover, this paper divides its sample into 80% training data and 20% test sample, and uses 25% of the training data as validation data to determine the optimal hyperparameters of the neural network models

Furthermore, this study employs robustness tests to test the findings from the main sample. These three robustness tests are (i) an alternative training set size for the main sample, (ii) similar multi-step forecast sizes ($H=4$ and $H=6$), and (iii) stocks from other geographic regions than the main sample. Needless to say, a new sample was used for the third robustness test. This sample consists of the following European stock indexes: Financial Times Stock Exchange 100 (FTSE), DAX Performance (GDAXI), and STXE 600 (STOXX). It is noteworthy that the sample period for these stock indexes is a bit shorter than the main sample, ranging from 25 May, 2004 until 2 March, 2022.

The main sample results indicate that NBEATSx-VP yields statistically significantly more accurate forecasts than the other considered models. The proposed model achieves at least 9% more precise forecasts than the other models. Additionally, the null hypotheses of both statistical tests are rejected for virtually all stock indexes and forecasting models considering the threshold of 0.01. Moreover, when using the threshold of 0.05, all null hypotheses are rejected. The superiority of NBEATSx-VP also holds in all robustness tests. While the first robustness test yields virtually the same results as the main sample, the superiority of NBEATSx-VP is a bit less present in the second robustness test when considering $H=6$. Consequently, it could be hypothesized that the superiority of NBEATSx-VP over other forecasting models vanishes as H increases. Nevertheless, more research is needed to test such a hypothesis and find a scientific explanation for the phenomenon. Lastly, the superiority of NBEATSx-VP over the other studied models is also clear in the third robustness test. Apart from NBEATSx-VP versus NBEATSx-V for STOXX, NBEATSx-VP produces statistically significantly more accurate forecasts than the other models for all stock indexes.

In conclusion, given the results of the main sample and robustness tests, it can be confidently affirmed that the proposed model is superior to current benchmark models. As proposed in Section 3.3, this is presumably the case due to the phenomenon that 2-dimensional Wasserstein distances of stock log returns and VIX close prices are complementary to each other as the first captures long-term patterns of RV while the latter captures short-term patterns of RV.

Finally, interesting directions of further research is the investigation of the performance of NBEATSx-VP in one-step forecast, with longer-term multi-step forecasting ($H \geq 6$), and with stocks from developing countries. Therefore, the author of this paper would like to kindly invite the scientific community to embark on these unexplored areas and contribute to further advancements in this field of research.

7. Data and Code Availability

Data will be made available upon request. For code, see

<https://github.com/hugogobato/Application-of-Persistent-Homology-in-Forecasting-Realized-Volatility.git>.

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